

PHILIP LIN

746 Penfield Avenue, Havertown, PA 19083 • Cell: 610-609-6734 • Home: 610-446-0761 • philip.lin@villanova.edu

EDUCATION

VILLANOVA UNIVERSITY

Villanova, PA

M.S. in Mechanical Engineering

Aug. 2010 – May 2011

- Five year Combined Bachelor's and Master's Program with Thesis; current GPA: 3.7

B.S. in Mechanical Engineering

Aug. 2006 – May 2010

- Concentration: Honors; Minors: Electro-Mechanical Engineering, Mathematics, and Chinese; Magna Cum Laude, GPA: Cumulative 3.9, Technical 3.9

INTERNSHIPS

GE ENERGY

Atlanta, GA

Energy Services Intern, Monitoring and Diagnostics Group

Summer 2010

- Reviewed and evaluated gas turbine combustion system monitoring (CBSM) algorithm by reviewing exhaust temperature and combustion can pressure dynamics for trends using statistical tools to filter out noise in the data
- Provided detailed next step recommendations on how to improve the CBSM algorithm to reduce false alarm rate and improve probability of detection

GE-HITACHI NUCLEAR ENERGY

Wilmington, NC

New Plants Engineering Intern, ESBWR Group

Summer 2009

- Performed water level and pressure sensitivity study for LOCA using nuclear plant modeling codes (TRACG)
- Supported licensing meetings and presentations to the NRC on topics such as non-condensable gas concentration, passive containment cooling system, and heat transfer across containment surfaces

EXELON NUCLEAR

Pottstown, PA

On-site Nondestructive Testing (NDT) Intern at Limerick Nuclear Power Plant

Summer 2008

- Supported Level II and III NDT specialists with emergency examinations in the field (inside plant and outdoor underground pits) when the NDT was short staffed, worked several 12 hour backshifts (6pm to 6am)
- Discovered inaccurate NDT procedures of scanning through corrosion to negate the need for costly emergency piping repairs (several hundred-thousand dollars), pushed idea through upper management, received recognition from project leader and acknowledgement from VP
- Managed two NDT GE contractors for a period of week when all the on-site Exelon NDT personal had taken vacation, acted as NDT point man for group involvement in emergency repair project
- Proactively identified and completed more technical side projects in ISFSI dry storage canister vacuum drying and moisture carryover monitoring of steam dryer

RESEARCH AND PUBLICATIONS

LABORATORY FOR ADVANCED THERMODYNAMICS AND FLUID SYSTEMS

Villanova, PA

Graduate Research Assistant for Dr. Alfonso Ortega

Fall 2010 – Spring 2011

- Research focus on spray cooling and single droplet impingement

Undergraduate Research Assistant for Dr. Alfonso Ortega

Fall 2007 – Spring 2010

- Awarded Undergraduate Research funding of \$2000 to study thermal response of a droplet impinging on a heated surface
- Presented findings in the National Conference in Undergraduate Research (NCUR) 2010 in Missoula, MT and published paper under in the *NCUR 2010 Proceedings* under the title "An Assessment of the Feasibility of Measuring the Transient Response of a droplet Impinging onto a Heated Surface using Micro-Thermocouples"
- Modified existing apparatus to perform ASTM "Standard Test Method for Thermal Conductivity by Means of the Guarded-Comparative-Longitudinal Heat Flow Technique" (Document E 1225-04)
- Retrofitted outdated experiment apparatuses with machine shop work

COMMUNITY INVOLVEMENT

THE CHURCH IN PHILADELPHIA

Philadelphia, PA

Youth Group Teaching Assistant

Fall 2005 – Present

- Help organize retreats for about 25 middle school to high school age Christians
- Teach at a Northeast regional Bible Summer School lasting 2 weeks
- Organize logistics and lead fellowship for about 15 young people every Saturday night

SKILLS

Language: Chinese (Mandarin Dialect), fluent in conversation, proficient in reading and writing

Technical: Ansys, Fluent, Labview, Matlab, Mathcad, Solidworks